1 Life-Cycle Model With Endogenous Labor Supply

- State variables:
  - a, assets.
  - η, transitory labor productivity shock.
  - j, age.
  - θ, permanent productivity type.
- Choice variables:
  - a', next-period assets.
  - l, labor supply, or hours of work.
- Household optimization problem in recursive form:

$$V(a,\eta,j,\theta) = \max_{a',l} \left\{ u(c, 1-l) + \beta\psi_{j+1} E\left[V(a',\eta',j+1,\theta) \mid \eta \right] \right\}.$$

The budget constraint is

$$c + a' = (1+r)a + we_j\theta\eta l + pen_j,$$

with

$$a' \geq 0, \quad 0 \leq l \leq 1, \quad c \geq 0.$$

Given a', labor supply has the following closed-form expression. Define

$$available = (1+r)a + pen_j, \quad wage = we_j\theta\eta.$$

For working ages, $j < J_R$,

$$\tilde{l}(a',a,\eta,j,\theta) = \nu + (1-\nu)\frac{a' - available}{wage},$$

and the constrained labor supply is

$$l(a',a,\eta,j,\theta) = \min\left\{ \max\left\{ \tilde{l}(a',a,\eta,j,\theta), 0 \right\}, 1 - 10^{-10} \right\}.$$

For retirement ages, $j \geq J_R$,

$$l(a',a,\eta,j,\theta) = 0.$$

The policy functions are

$$\begin{aligned} a' &= g_a(a, \eta, j, \theta), \\ l &= g_l(a, \eta, j, \theta). \end{aligned}$$

Effective labor productivity is

$$h_j(\eta, \theta) = \begin{cases} e_j \theta \eta, & j < J_R, \\ 0, & j \geq J_R, \end{cases}$$

where e_j is the deterministic age-efficiency profile. In the Fortran code, $r = 0.04$, $w = 1$, $J = 80$, $J_R = 45$, $N_a = 200$, $N_\theta = 2$, and $N_\eta = 7$. The pension is

$$pen_j = \begin{cases} 0, & j < J_R, \\ \kappa \tau_p \dfrac{1}{J_R - 1} \sum_{s=1}^{J_R - 1} e_s, & j \geq J_R, \end{cases}$$

where $\kappa = 0.5$ is the replacement parameter and $\tau_p = 0.33$ is the pension contribution parameter. The period utility function is

$$u(c, 1 - l) = \frac{\left[c^\nu (1 - l)^{1-\nu}\right]^{1 - \frac{1}{\gamma}}}{1 - \frac{1}{\gamma}},$$

with $\nu = 0.335$, $\gamma = 0.50$, and $\beta = 0.98$.

The transitory productivity process is discretized as an AR(1) process,

$$z' = \rho z + \varepsilon', \quad \varepsilon' \sim N(0, \sigma_\varepsilon),$$

with $\rho = 0.985$ and innovation variance $\sigma_\varepsilon = 0.022$. The corresponding innovation standard deviation is $\sqrt{0.022}$. The code stores $\eta = \exp(z)$ after discretization. Let

$$\pi(\eta, \eta') = \Pr(\eta_{j+1} = \eta' \mid \eta_j = \eta)$$

denote the transition probability induced by the discretized Markov chain. The two permanent productivity types are

$$\theta_1 = \exp(-\sqrt{\sigma_\theta}), \quad \theta_2 = \exp(\sqrt{\sigma_\theta}),$$

with $\sigma_\theta = 0.242$ and equal type probabilities.

Initial condition: at age 1, all individuals start with zero assets and the middle transitory shock state, which corresponds to $z = 0$ and hence $\eta = 1$. The permanent productivity type is distributed uniformly across the two permanent types.

- Distribution:

$$\mu(a_k, \eta', j+1, \theta) = \sum_a \sum_\eta q_k(g_a(a, \eta, j, \theta))\pi(\eta, \eta')\mu(a, \eta, j, \theta),$$

  for $j = 1, \ldots, J-1$, where $q_k(\cdot)$ is the interpolation weight on asset grid point a_k. If $g_a(a, \eta, j, \theta)$ is not a grid point, the code linearly redistributes mass across the two neighboring asset-grid points. Let $a_k \leq g_a(a, \eta, j, \theta) < a_{k+1}$ and choose ω such that

$$g_a(a, \eta, j, \theta) = \omega a_k + (1-\omega)a_{k+1}.$$

  Then $q_k = \omega$, $q_{k+1} = 1 - \omega$, and all other asset-grid weights are zero.

  The initial distribution is

$$\mu(a, \eta, 1, \theta_i) = \begin{cases} \pi_i^\theta, & a = 0,\ \eta = 1,\ \theta = \theta_i, \\ 0, & \text{otherwise.} \end{cases}$$

2 Mapping Into VFI-Toolkit Notation

In VFI-Toolkit notation, θ is a permanent type and should be handled through the permanent-type dimension. The shock η is the exogenous Markov shock, corresponding to z in the toolkit notation. Assets a are the endogenous state, while a' is the next-period asset choice.

Labor l is conceptually a decision variable, so it corresponds to d in the toolkit notation. However, because the closed-form solution gives l as a function of a', a, η, j, and θ, we do not need to declare a separate d grid. The return function can instead compute labor internally and use $(aprime, a, z)$ as the relevant action-state inputs, with age j handled through the finite-horizon setup.

3 Run-Time Comparison

Step	Fortran (s)	MATLAB (s)
solve_household (VFI)	0.182225	0.896763
distribution	0.013436	0.077600
aggregation (moments)	0.001490	2.965766
total	0.197152	3.940129

The MATLAB timings were obtained from the VFI-Toolkit implementation in `codes_matlab/main.m`. The reported MATLAB total is the sum of the value-function iteration, distribution, and model-moment evaluation steps; it does not include writing the output table or plotting.

The Fortran code was compiled with Intel oneAPI `ifx` using the makefile option string `/warn:all /Qopenmp /O2`. The computation is serial: there are no OpenMP parallel directives in the model code. The speed-relevant option is `/O2`, which enables compiler optimization. The `/Qopenmp` option is used to enable OpenMP runtime support for `omp_get_wtime()`, the timer used for the runtime measurements. The `/warn:all` option is diagnostic rather than performance-oriented.